COMP 3500SEF · Final Report

Warehouse Logistics Management System
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Testing: Oscar Chung Ngon Man, Li Mu Qiu.

1. Scope

Goals: traceable decisions (each order's history auditable via ledger and notifications) and role-separated workflows. Local-only: one app; seed data (RK-10).

2. Development process

Table A. Increments

Increment and date	Trigger	Outcome
Planning, 9–11 Oct 2026	Coursework kickoff	Scope, responsibilities and API contracts recorded (register)
17 Oct review	Stock and order review	Two defects; fixed as row 3
Confirmation, 26–28 Oct 2026	Row-2 defects: repeat submits, no ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Role functions integrated	All four role workflows (§1) run end to end
Driver and privacy, 2 Nov 2026	Driver tests outstanding	Both pass (driver, privacy)
Diagram review, 10–11 Nov 2026	Diagrams diverged	Aligned (review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date export in three interactions (fill both dates, export)
CI health tests, 11–12 Nov 2026	No test covered CI-run outcomes	Two regression tests; suite reached 38 (unit.log)

3. Requirements

- FR-01 create orders with lines Test. - FR-02 enforce lifecycle, stock release Test (cancel). - FR-03 number orders, shipments, owners Test. - FR-04 inventory, ledger, low-stock alerts Test. - FR-05 record, read notifications Test. - FR-06 admin aggregates, CSV Test. - FR-07 authenticate, access Test. - FR-08 assign, deliver Test.

- NFR-01 performance: API p95 ≤2 s, first contentful paint p95 ≤3 s on the local host (measurement). - NFR-02 CI health — every commit ends in green unit, integration, browser runs (runs, observations). - NFR-03 least privilege: role gates; delivery address only to the assigned driver (privacy tests). - NFR-04 interaction budget: every core operation (the eleven in §8) takes at most three in-app interactions; one is a fill, selection or click (walkthrough). - NFR-05 process quality — every increment closes on a recorded review or test run.

Table B. Risk coverage

Risk	Addressed by	Evidence
①RK-01, RK-03, RK-06, RK-08 schedule, test week, coursework, video (recorded)	weekly re-planning, scope priorities	risk plan
②RK-02 contract rework	lifecycle locked as test constants	constants tests
③RK-04, RK-07 setup, stack	setup guide; CI per commit	README ci.yml
④RK-05 demo data	deterministic seed, regression-tested	seed test
⑤RK-09, RK-10 hosting; personal data	accepted — seed data only (§1); NFR-03 still applies	risk plan

4. System models

Order items reference one product; orders belong to one user; ≤1 shipment records events; notifications relate to orders.

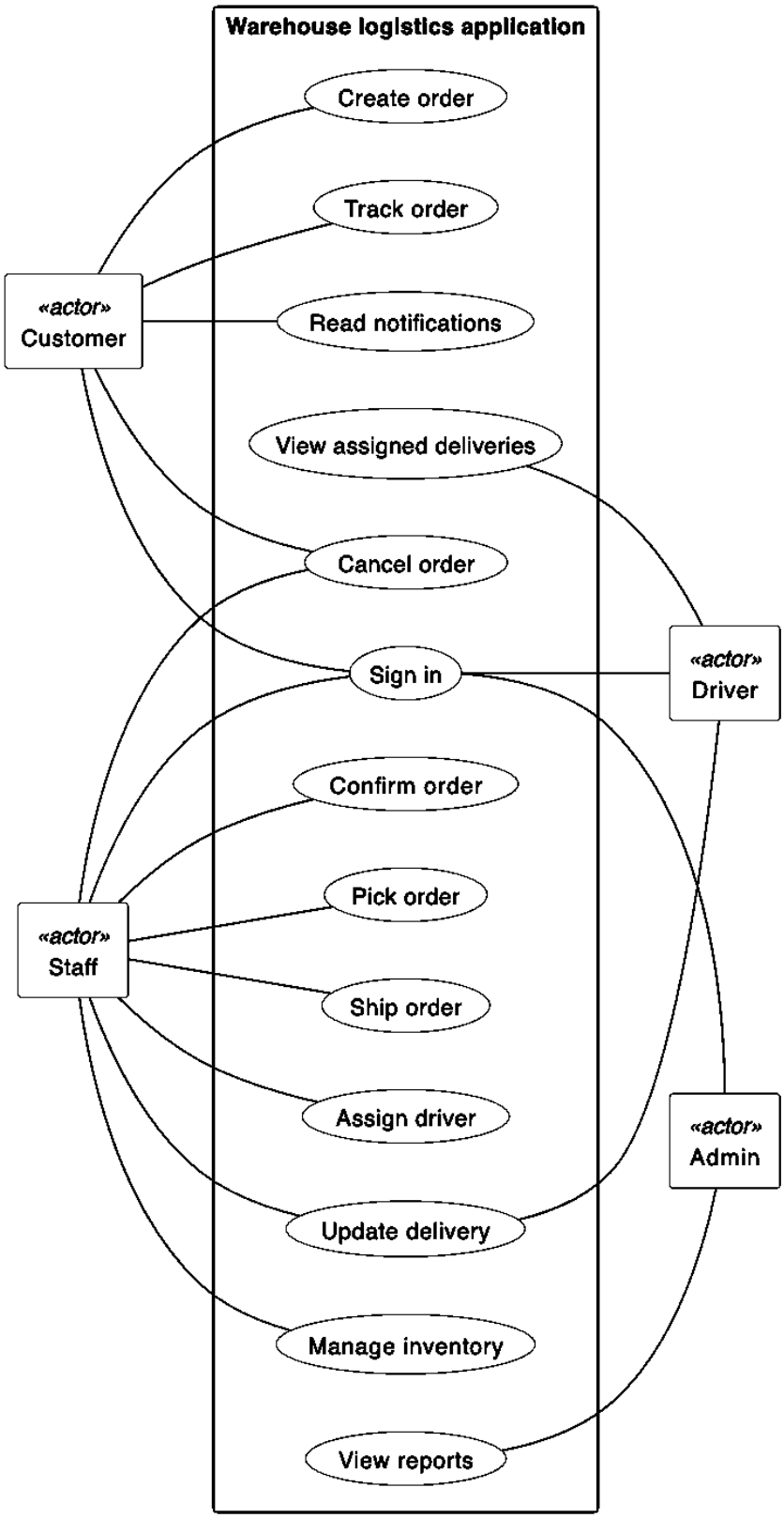


Figure 1.

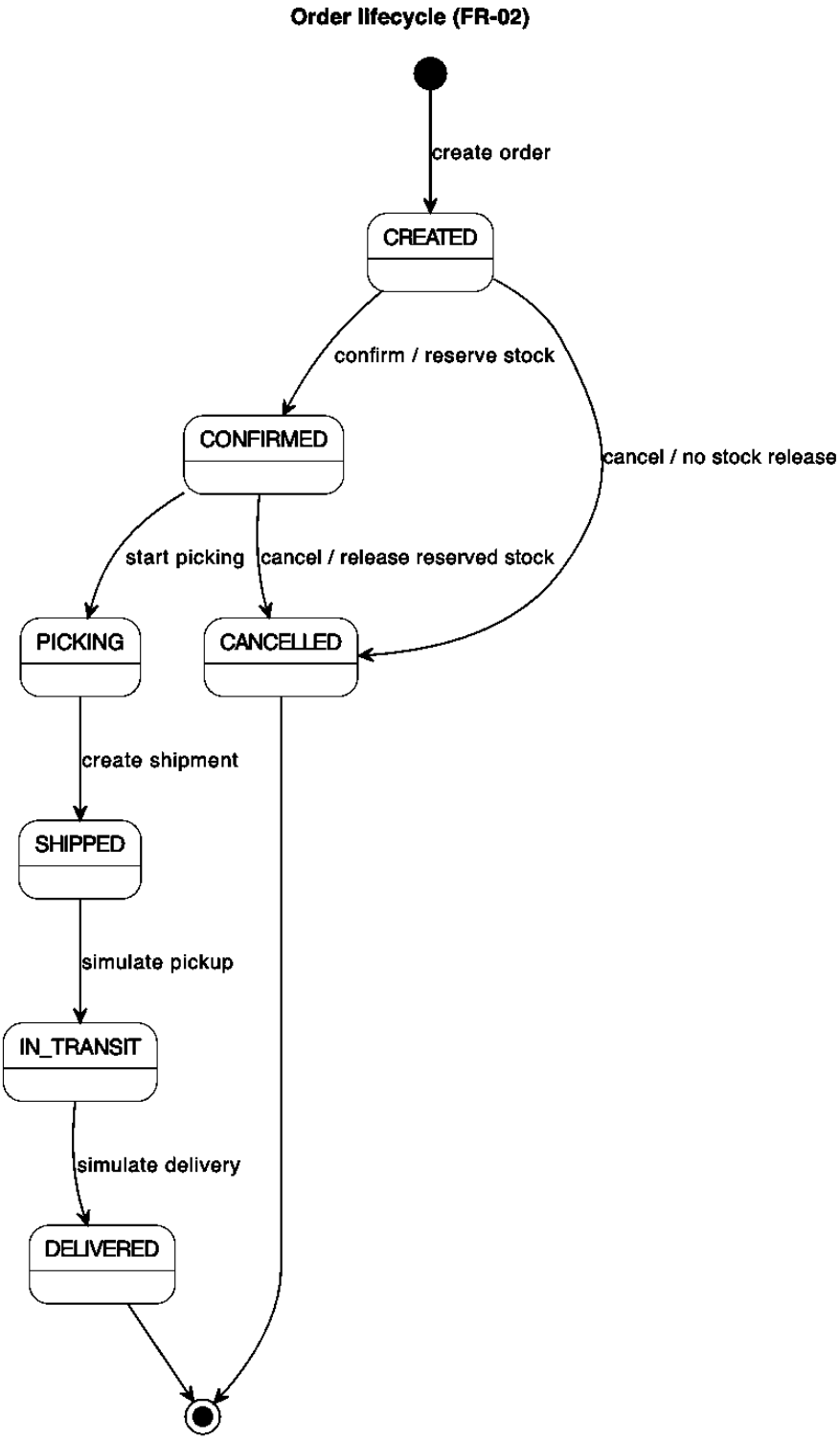


Figure 2. Order transitions: reserve, release, or unchanged.

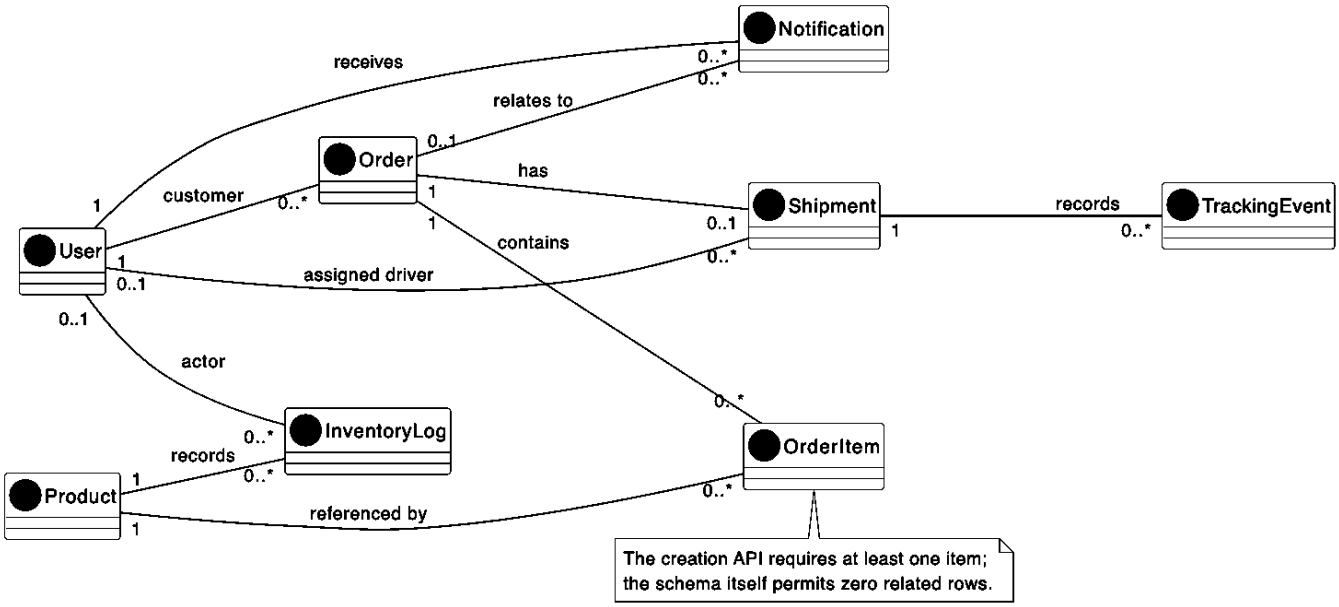


Figure 3.

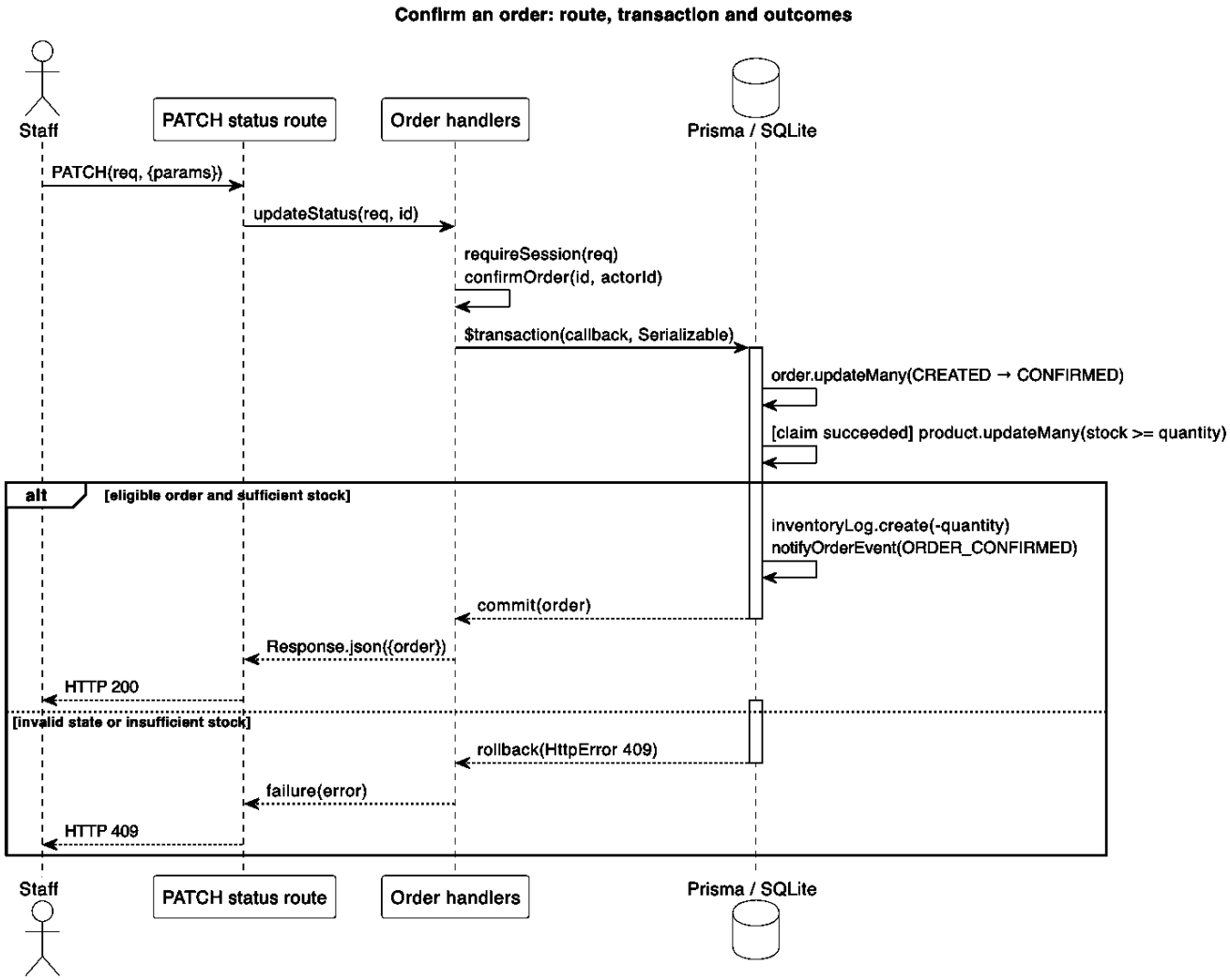


Figure 4. Confirmation: guard, rollback.

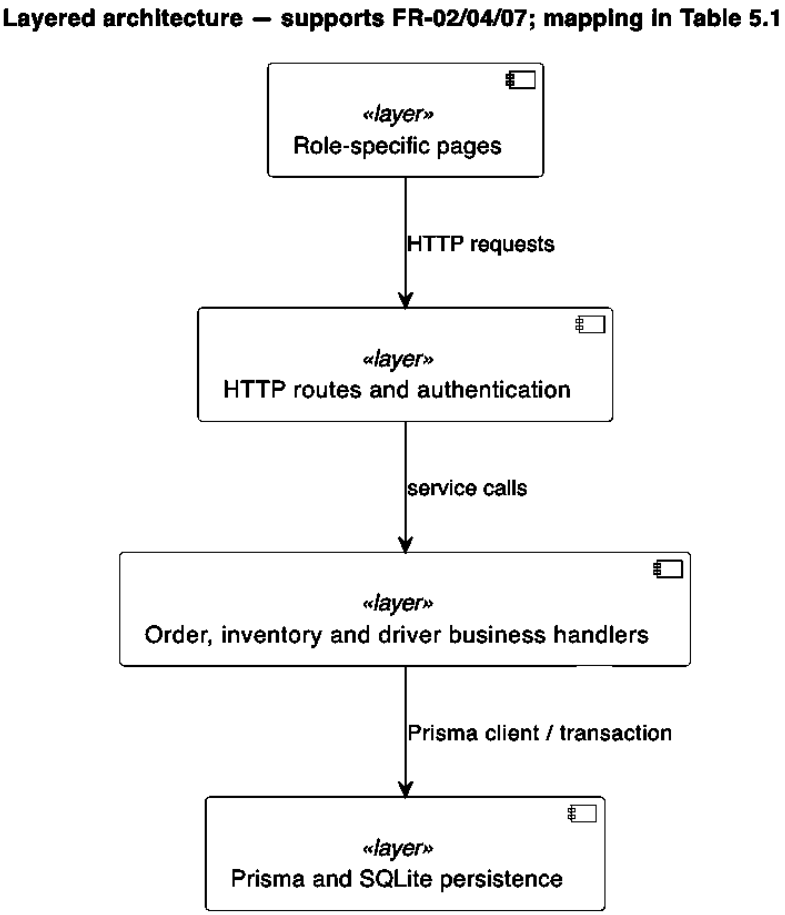


Figure 5.

5. Implementation

Confirmation reserves stock; lifecycle CREATED→CONFIRMED→PICKING→SHIPPED→IN_TRANSIT→DELIVERED; cancel from CREATED or CONFIRMED.

Cancellation permissions:

Role	Transition and effect
Customer	CREATED: cancel, no restock
Customer	CONFIRMED: cancel, restock
Staff	CREATED: rejected
Staff	CONFIRMED: cancel, restock
Customer, staff	PICKING onwards: rejected

A CREATED order awaits the customer's cancel or the staff confirmation; staff have no refuse action.

Atomic confirmation (HTTP 409 on repeats; rollback on short stock); driverId set at SHIPPED, immutable after; FR-03/04 verified (order-api, driver-api, inventory-api).

6. Verification

Verification checks the implementation against §3; validation exercises the delivered workflows (§8). CI records 38 unit, 140 integration, 2 integrity checks: the demo-data reset requires RESET_DEMO_DATA=YES (evidence protection), and the packaged submission's SHA-256 manifest must match the source (integrity checks). Logs: integration "tests 140".

Loopback measurement: p95 3.2–13.6 ms, first contentful paint p95 64–96 ms (30 requests × 6 endpoints) vs 2/3 s — loopback, not representative of public networks.

Three reviews (rows 2, 6, 7): rows 2 and 6 verified requirements and models; row 7's fix was validated through the interface (§8).

7. Evolution

Three reviews: 17 Oct found two defects (repeat submits, missing ledger) fixed by §5; 10–11 Nov aligned diagrams and NFR-01/02 to CI-run checks; 11 Nov added direct export (issue #15). Untested CI record added two regression tests (row 8; 36→38).

8. Validation

Walkthrough covers eleven operations (Figure 1): create order; open details; confirm; start picking; ship; assign driver; record pickup; record delivery; filter report; export current; export selected dates; ≤3 interactions each. "Browser acceptance passed: 26 checks": 20 checks (five operations once, three twice, three thrice) plus 6 cross-cutting (two each: validation, session, interface). During report preparation, 2 classmates each attempted 32 core tasks: 64 of 64 attempts succeeded unaided, scored by the two recorders Oscar Chung Ngon Man and Li Mu Qiu; median 2.1 min (user-validation).

9. Feedback and reconciliation

Sources: whole-team reviews; test runs and the scripted walkthrough executed by Oscar Chung Ngon Man and Li Mu Qiu (Testing).

Table C. Feedback

Feedback (source, date)	Disposition and evidence
Diagrams diverged (10–11 Nov, corrections)	Corrected
Stock (FR-04) (17 Oct, traceability)	Moved to FR-02
Thin triggers (10–11 Nov, register)	Rewritten
Filter-export: four interactions (11 Nov, issue #15)	Combined path rejected — its steps already totalled four; direct path meets NFR-04; issue closed
No CI-health test (11 Nov, unit.log)	Two regression tests (row 8); suite 38
NFR-01/02 assumed hosted measurement, daily window (10–11 Nov, revision log)	CI-run checks; loopback

10. Evidence

38 unit, 140 integration, 2 integrity from the 12 Nov run (the only additions since the 11 Nov browser run were tests); 26 browser checks (11 Nov) — BASELINE. Traceability.